LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	31
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Brief explanation of ILS system, ILS coupling and VOR system and VOR coupling
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Brief explanation of ILS system, ILS coupling and VOR system and VOR coupling.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of AFCS coupling logic diagrams; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary navigation systems used for approach?
(b) Define the function of the AFCS in flight control.
(c) List the two main components of the VOR system.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• ILS Localizer Function • ILS Glideslope Function • ILS Coupling Mechanism • AFCS ILS Integration	L & Ds	PPT, IB	15 min
(b)	• VOR System Overview • VOR Coupling Process • AFCS VOR Integration • Coupling Modes Comparison	L & Ds	PPT, PDF	20 min
(c)	• ILS vs VOR Coupling • AFCS Automation Levels • System Failure Indications • Manual Override Procedures	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the AFCS couples the ILS localizer signal.
(b) Describe the difference between VOR and ILS coupling in the AFCS.
(c) Identify the indications of an ILS coupling failure.

12. Home Assignment and Follow Up	Prepare a brief note on the integration of the Inertial Navigation System with the MHRS system as covered in LP 32.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Explanation of inertial navigation system and integration with MHRS system.	Refer to the master syllabus reference list	LP 32

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min